INTRODUCTION TO IML F26 PROJECT: SYMMETRIC FIXED POINTS OF THE RICCI FLOW

1. Vague introduction of some geometric background

Roughly speaking, the Ricci flow evolves curved surfaces, which in higher dimensions are objects called manifolds, in ways which have been very useful to mathematical applications (eg. the 3D Poincaré conjecture).

Explicitly, given a Riemannian manifold (M^n, g_0), where

$$M^n \text{ is an } n\text{-manifold} \qquad \text{and} \qquad g_0 \text{ is a Riemannian metric,}$$

a Ricci flow starting from (M^n, g_0) is a family of Riemannian metrics $\{g(t)\}_{t\in[0,T)}$ for some $T > 0$ satisfying the evolution equation

$$\begin{aligned} \frac{\partial}{\partial t} g(t) &= -2\mathrm{Ric}(g(t)), \\ g(0) &= g_0. \end{aligned}$$

Above, $\mathrm{Ric}(g(t))$ is a notion of curvature which is calculated given an input Riemannian metric $g(t)$. A Riemannian metric completely determines distances and notions of curvature on M^n.

Here is an example: Let (S^2, g_0) denote the round unit sphere in $\mathbb{R}^3$, with g_0 the standard round metric induced by the Euclidean metric $dx^2 + dy^2 + dz^2$ in $\mathbb{R}^3$. It has radius 1 and circumference 2π. Then the Ricci flow starting from $(S^2, g(t))_{t\in[0,1/2)}$ is given by time-slices $(S^2, g(t)$ which are round spheres of radius $\sqrt{1-2t}$ with circumference $2\pi\sqrt{1-2t}$.

It is easier to talk about this at the board so I will discuss more background and draw some associated pictures on Friday, 9/4.

2. What we want to do, explicitly

We want to understand some geometric properties of self-similar solutions of the Ricci flow $(M^n, g(t))_{t\in[0,1)}$. This means that each time-slice $(M^n, g(t))$ looks the "same" as any other time-slice, modulo scaling (like the shrinking sphere example in Section 1). These self-similar solutions are called Ricci solitons. Since each time-slice of a Ricci soliton is the same as any other time-slice up to scaling, we often just refer to an expanding Ricci soliton by a single time-slice (M, g).

There are three families of Ricci solitons which we'd like to study, to start out with. Two families, are on $\mathbb{R}^3 \times S^1$ and $\mathbb{R}^2 \times S^2$, from [NW24, Raj25]. The third family is on $\mathbb{R}^4$ (or $\mathbb{R}^3$), and due to Bryant [Bry05] (see also [Hsu21] and [Cho14, Appendix A]).

2.1. **Ricci solitons on $\mathbb{R}^3 \times S^1$ and $\mathbb{R}^2 \times S^2$.** We use the parametrizations from [Raj25] since these are easier to understand. Abstractly, we are looking for $SO(3) \times SO(2)$-invariant Ricci solitons given by (M, g) and a function $f : M \to \mathbb{R}$ (where M is $\mathbb{R}^3 \times S^1$ or $\mathbb{R}^2 \times S^2$) satisfying the equation

$$\mathrm{Ric} + \nabla^2 f + g = 0.$$

By the imposed $SO(3) \times SO(2)$ symmetry, we can write

$$g = dr^2 + a(r)^2 g_{S^1} + b(r)^2 g_{S^2},$$

where $r \in [0, \infty)$ is the radial coordinate on $\mathbb{R}^3$ or $\mathbb{R}^2$, and also take $f : [0, \infty) \to \mathbb{R}$ to depend only on the radial coordinate as well. After some geometric computations, this means we need to study solutions of the following ODE system (see [Raj25, §2]):

$$\begin{aligned} f'' &= \frac{a''}{a} + 2\frac{b''}{b} - 1, \\ a'' &= -2\frac{a'b'}{b} + a'f' + a, \\ b'' &= \frac{1 - (b')^2}{b} - \frac{a'b'}{a} + b'f' + b, \end{aligned} \tag{2.1}$$

for functions

$$a : [0, \infty) \to [0, \infty), \quad b : [0, \infty) \to [0, \infty), \quad f : [0, \infty) \to \mathbb{R}$$

subject to either the initial conditions

$$\begin{gathered} a(0) = a_0 > 0, \quad a'(0) = 0, \quad b(0) = 0, \quad b'(0) = 1, \\ f(0) = f'(0) = 0, \quad f''(0) = f_0 < 0, \end{gathered} \tag{2.2}$$

in the case where $M = \mathbb{R}^3 \times S^1$, or

$$\begin{gathered} a(0) = 0, \quad a'(0) = 1, \quad b(0) = b_0 > 0, \quad b'(0) = 0, \\ f(0) = f'(0) = 0, \quad f''(0) = f_0 < 0, \end{gathered} \tag{2.3}$$

in the case where $M = \mathbb{R}^2 \times S^2$.

It turns out this ODE system is always uniquely solvable:

Theorem 2.1 ([Raj25, Theorem 5.1])**.** *For any set of initial data $(a_0, f_0) \in \mathbb{R}_{>0} \times \mathbb{R}_{<0}$ in the case $M = \mathbb{R}^3 \times S^1$, or $(b_0, f_0) \in \mathbb{R}_{>0} \times \mathbb{R}_{<0}$ in the case $M = \mathbb{R}^2 \times S^2$, there exists a unique solution of the ODE system (2.1).*

However, there are a number of questions which would be of some research interest which we can try to study/answer (stated vaguely; any concrete information would be nice, and see [Raj25, §6–8] for some related discussions):

Question 1. Given a solution (a, b, f) of (2.1), what can we say about $a'_\infty := \lim_{r\to\infty} a'(r)$ and $b'_\infty := \lim_{r\to\infty} b'(r)$? These limits are known to be positive real numbers which describe the geometric "asymptotic cone" of the solutions. Or more concretely, how do the limits a'_∞, b'_∞ relate to the initial data (a_0, f_0) or (b_0, f_0)?

Question 2. There should be some limits (a'_∞, b'_∞) achieved by multiple pairs of initial data. Can we say what these are, or find some subset of $\mathbb{R} > 0 \times \mathbb{R} < 0$ which is guaranteed to have one of these limits?

One curvature quantity that would be interesting to know more about is the **scalar curvature** of the metric. The scalar curvature R at each point (a number) is calculated by the formula:

$$R = -2\frac{a''}{a} - 4\frac{b''}{b} - 3\frac{a'b'}{ab}.$$

2.2. **Ricci solitons on $\mathbb{R}^4$ (or $\mathbb{R}^n$ in general).** Similar to in Section 2.1, the Ricci soliton equations will now give an ODE system when we look for $SO(n)$-symmetric solitons on $\mathbb{R}^n$, but this time we will only have two functions, by [Bry05]. We can consider similar geometric questions as well.

This time, we can compute our soliton equation with respect to the metric

$$g = \frac{da^2}{h(a^2)} + a^2 dg_{S^{n-1}},$$

where $h : [0, \infty) \to [0, \infty)$ solves (see [Bry05, 2.9] and [Hsu21, (1.6)]; we are interested in the case in which $\lambda = 1$):

$$2r^2 h(r)h''(r) = h(r)(h(r) - 1) + rh'(r)(rh'(r) - r - 1), \tag{2.4}$$

subject to the initial conditions

$$h(0) = 1, \quad h'(0) = \mu_1 \in \mathbb{R}.$$

Once one has solved for h, it is easy to solve for f [Bry05, (2.10)].

This time, we are interested in the quantity $\lim_{r\to\infty} \sqrt{h(r)}$, which again is related to the "asymptotic cone" of the associated Ricci soliton.

Question 3. Can we give any more information on the set below?

$$\{ \lim_{r\to\infty} \sqrt{h(r)} : \ h \text{ solves } (2.4)\}$$

For example, by [Hsu21, Theorem 1.8], if $\mu_1 < 0$ then $\lim_{r\to\infty} \sqrt{h(r)} \in (0, 1)$, and in general $\lim_{r\to\infty} \sqrt{h(r)}$ is a finite number only when $\mu_1 < \frac{1}{n}$.

Again, the behavior of the scalar curvature at each point of $(\mathbb{R}^n, g)$ is of interest. This time, the formula for R is

$$R = -2(n-1)\frac{a''}{a} + (n-1)(n-2)\frac{1 - (a')^2}{a^2},$$

where a is the function which satisfies $a'(t) = \sqrt{h(a(t)^2)}$.

References

[Bry05] Robert L. Bryant. Ricci flow solitons in dimension three with $SO(3)$-symmetries. Preprint, 2005.

[Cho14] Otis Chodosh. Expanding Ricci solitons asymptotic to cones. *Calc. Var. Partial Differential Equations*, 51(1-2):1–15, 2014.

[Hsu21] Shu-Yu Hsu. A new proof for the existence of rotationally symmetric gradient Ricci solitons. *arXiv e-prints*, page arXiv:2105.03805, May 2021.

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